

Functional sequences, series and spaces – 3

Power series

Homework

13 сентября 2026 года

Theoretical minimum. The radius of convergence is given by $1/R = \limsup |a_n|^{1/n}$ and survives differentiation and integration; inside the interval the sum is C^∞ and everything may be done term by term, on the boundary nothing is automatic. Abel's theorem says that convergence at the endpoint forces the radial limit to agree with the sum; the converse requires a Tauberian condition, $a_n = o(1/n)$ for Tauber and $a_n = O(1/n)$ for Littlewood. Cesàro summability implies Abel summability but not conversely, so the Abel method is strictly stronger. Smoothness does not imply analyticity: the criterion is $|f^{(n)}| \leq C^n n!$, and non-negativity of all derivatives is enough as well. On the boundary of the disc there is always a singularity, at $x = R$ when the coefficients are non-negative; a series with large gaps is singular everywhere on the boundary. Finally, integer coefficients are a strong restriction: they cannot tend to zero without vanishing.

1. Prove that the Cauchy product of two power series is again a power series with at least the smaller radius, deduce $(\sum \frac{x^n}{n!})^2 = \sum \frac{2^n x^n}{n!}$, and recall from Series and infinite products – 5 an example of two convergent numerical series whose Cauchy product diverges.
2. (Proposed problems) Let R be a commutative ring of characteristic 2 and consider in $R[[X]]$ the formal power series $f(X) = \sum_{n \geq 0} X^{2^n - 1}$. Prove that for all $\alpha, \beta \in R$ the series $f(\alpha^2 X) f(\beta^2 X) f((\alpha + \beta)^2 X)$ is a square in $R[[X]]$.

3. Find the radius of convergence of $\sum d(n)x^n$, $\sum \sigma(n)x^n$ and $\sum p(n)x^n$, where $p(n)$ is the number of partitions, and determine the order of growth of each sum as $x \rightarrow 1^-$.
4. Compute the Abel sums of $\sum (-1)^n$ and $\sum (-1)^n n$, and prove that Abel's method sums everything that Cesàro's method sums, with a series that is Abel summable but not Cesàro summable.
5. Prove that the unit circle is a natural boundary for $\sum_{n \geq 0} x^{2^n}$: show that the radial limit is infinite at every dyadic root of unity and deduce that the sum has no analytic continuation across any arc.
6. Prove Pringsheim's statement in real form: if $a_n \geq 0$, the radius equals 1 and $\sum a_n = \infty$, then $f(x) \rightarrow \infty$ as $x \rightarrow 1^-$; and give an example with $a_n \geq 0$, radius 1 and f bounded on $[0, 1)$.
7. Prove that a formal power series with $a_1 \neq 0$ has a formal inverse under composition, compute the first three coefficients of the inverse of $x + x^2$, and state the Lagrange inversion formula.
8. Prove the uniqueness theorem: if $\sum a_n x^n = 0$ on a set having an accumulation point inside the interval of convergence, then all $a_n = 0$. Deduce that two power series agreeing on a sequence $x_k \rightarrow 0$, $x_k \neq 0$, coincide.
9. Using generating functions, find closed forms for the Fibonacci and the Catalan numbers, and prove Euler's identity that the number of partitions into distinct parts equals the number of partitions into odd parts.
10. Prove that a power series with integer coefficients and radius of convergence greater than 1 is a polynomial, and decide what can be said when the radius equals 1.